



Lab 08

Z-Transform

Name:	Student ID:
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8.1 Objective

To investigate the properties of the Z-transform and its application in analyzing Discrete-Time (DT) Linear Time-Invariant (LTI) systems.

8.2 Introduction

The Z-transform is a mathematical tool used in signal processing to analyze discrete-time signals in the frequency domain. It converts signals from the time domain into the Z-domain, making it easier to perform various operations like convolution, filtering, and stability analysis. It simplifies working with digital signals by analyzing them in the frequency domain instead of the time domain. This is useful in various applications such as signal detection, system identification, and filter design. It is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

For an LTI system characterized by a difference equation, the Transfer Function $H(z)$ relates the input $X(z)$ to the output $Y(z)$. System stability is determined by the Region of Convergence (ROC) and the location of poles relative to the unit circle.

Task 1: Computing Z-Transforms in MATLAB

Steps

1. Use MATLAB symbolic tools to compute the Z-transform of the following discrete-time signals:
 - a) $x_1[n] = u[n]$
 - b) $x_2[n] = 0.5^n u[n]$
 - c) $x_3[n] = \delta[n]$
2. Use the MATLAB commands `ztrans()` and `iztrans()` to compute the transforms.

Task 2: The Eigenfunction Property

Theory: If the input to an LTI system is a complex exponential z^n , the output is the same signal scaled by the transfer function $H(z)$.

**Steps:**

1. Define a system $y[n] - 0.5y[n-1] = x[n]$. Using the Z-transform, derive the system transfer function $H(z)$. Taking Z-transform of both sides:

$$Y(z) - 0.5z^{-1}Y(z) = X(z)$$

Hence,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1-0.5z^{-1}}$$

2. Apply input $x[n] = e^{j(\frac{\pi}{4})n}$ and compute system output $y[n]$ using filter command.
3. Verify that $y[n] = H(e^{j(\frac{\pi}{4})n}) \cdot x[n]$
4. Attach the plot showing the real part of the filtered output compared to the theoretical result, plot of the ratio $|y[n]/x[n]|$ showing its convergence to the magnitude of $H(z)$ and calculated value of $H(e^{j(\frac{\pi}{4})n})$ in rectangular form.

Task 2: Iterative Time-Domain Analysis vs. Inverse Z-Transform

Solve a difference equation sample-by-sample using a loop and verify the result against the analytical formula derived from the Z-transform.

Steps:

1. Define an impulse input and implement the equation $y[n] = x[n] + 0.8y[n-1]$ using a for loop (Assume zero initial conditions).
2. Compute $y[n]$ for the first 10 samples using the difference equation.
3. Now solve the same system analytically using the Z-transform.
4. Apply the Z-transform to the difference equation. Derive the transfer function $H(z)$.
5. For an impulse input, compute $Y(z)$.
6. Use `iztrans()` to obtain the time-domain expression for $y[n]$.
7. Create a comparison table showing the first 10 values of:
 - $y[n]$ obtained from the **for-loop iteration**
 - $y[n]$ obtained from the **analytical expression**

Task 3: Stability Analysis via Pole-Zero Mapping

Theory: A causal LTI system is stable if all its poles lie inside the unit circle ($|p| < 1$).

Step:

1. Analyze **System A:** $H_1(z) = \frac{1}{1-0.8z^{-1}}$ and **System B:** $H_2(z) = \frac{1}{1-1.2z^{-1}}$
2. Use the command `zplane` to plot the poles and zeros of both systems.



3. Generate the impulse response using `impz`.
4. Apply a step input to both systems and observe their outputs in the time domain. Use the MATLAB command `stepz` to generate the step responses.
5. Attach the pole-zero maps for both systems, Impulse response plots for both systems, step responses and brief explanation of why System B is unstable based on the results.

Task 4: Practical Application – Multipath Signal in Wireless Communications

Theory: In urban environments, a signal $x[n]$ often reaches a receiver via multiple paths. This can be modeled as $y[n] = x[n] - \alpha x[n - N_0]$, where α is the attenuation factor and N_0 is the delay. Using the Z-transform, the system transfer function is $H(z) = 1 - \alpha z^{-N_0}$. To "fix" the system and recover $x[n]$, we must apply an inverse filter $G(z) = 1/H(z)$.

Step:

1. Create a random signal $x[n]$ and pass it through the multipath system $H(z)$ with $\alpha = 0.8$ and $N_0=11$.
2. Observe the corrupted signal $y[n]$ and notice how the reflections distort the original data.
3. Define the inverse system $G(z)$.
4. Pass the corrupted signal $y[n]$ through $G(z)$ using the filter command and compare the output with the original $x[n]$.
5. Attach the plots of original signal, corrupted multipath signal, and the successfully recovered signal.

**Assessment Rubric****Lab 08****Lab 8: Z-Transform**

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Marks distribution:

	Tasks	LR2 Code	LR5 Results	AR4 Report Submission	AR7 Comments
In-Lab	Task 1	/10	/10	/10	/10
	Task 2	/10	/10		
	Task 3	/10	/10		
	Task 4	/10	/10		
Total =100		/40	/40	/10	/10
CLO Mapped		CLO1	CLO1	CLO3	CLO3

CLO	Total Points	Points Obtained
1	80	
3	20	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	*Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
AR7	Report Content/Code comments	Most of the questions are not answered / figures are not labelled/ titles are not mentioned / units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are properly mentioned. Proper comments are stated in the code.